

Git & GitHub Handbook

A beginner-friendly guide to version control, branches, and merging workflows.

1. Git vs. GitHub

Though often used together, they serve completely different purposes:

Git (Local Engine)

- Installed locally on your computer.
- Manages and tracks changes in your files.
- Works completely offline.
- Allows you to revert to past versions of your code.

GitHub (Cloud Platform)

- A website hosting service for your Git repositories.
- Enables backing up code online.
- Allows collaboration with other developers.
- Provides visual tools like Pull Requests and Issues.

2. Understanding Branches

Imagine your codebase is a **tree**. The `main` branch is the tree trunk—it represents the live, production-ready website code.

A **branch** is a parallel stream of development. It is an exact copy of the trunk that grows out to the side. You can edit, test, or add features in this branch without changing or breaking the trunk.

Why we work in branches:

- **Safety:** If you write bad code that crashes the app, the live website remains unaffected.
- **Collaboration:** Two developers can work on different features on different branches without conflict.

- **Code Reviews:** Other developers can review your branch before it is brought into the main code.

3. The Git Workflow (Our Recent Session)

Here are the steps we executed to update the website's SEO service card to Social Media Marketing:

1. **Create and switch to a branch:**

```
git checkout -b feature/replace-seo-with-smm
```

2. **Modify files locally:** We edited `About.jsx` and `ContactPage.jsx` to replace SEO content.

3. **Stage the files:**

```
git add src/components/About.jsx src/components/ContactPage.jsx
```

4. **Commit (save) locally:**

```
git commit -m "Replace SEO and Content with Social Media Marketing"
```

5. **Push online to GitHub:**

```
git push -u origin feature/replace-seo-with-smm
```

4. Merging Your Branch to Main

Once you verify that your branch works perfectly, you merge it into the `main` branch:

Option A: The GitHub Way (Recommended)

Go to your GitHub repository online, click "**Create pull request**", review the diffs, and click the green "**Merge pull request**" button. Then update your local terminal:

```
git checkout main  
git pull
```

Option B: The Command Line Way

Merge the feature branch into `main` locally on your machine, then push the merged results to GitHub:

```
git checkout main  
git pull origin main  
git merge feature/replace-seo-with-smm  
git push origin main
```

5. Quick Cheat Sheet of Essential Commands

Command	Description
<code>git status</code>	Check current branch status and modified files.
<code>git checkout -b <branch></code>	Create a new branch and switch to it.
<code>git checkout <branch></code>	Switch to an existing branch (e.g. <code>git checkout main</code>).
<code>git add .</code>	Stage all modified files for committing.
<code>git commit -m "msg"</code>	Save staged changes locally with a descriptive message.
<code>git push</code>	Upload local commits to the online branch on GitHub.
<code>git pull</code>	Download the latest changes from GitHub to your computer.